11-3 Simulations [2]

Point!

1 Using Functions in Spreadsheet Software

(1) ([1]**Random number**) : A number that can appear with equal probability within a certain range.

(2) Functions of Spreadsheet Software

[1] ([2]**SUM (range1:range2)**) : Calculates the sum for cell range1 to range2.

[2] ([3]**IF (logical_expression, value_if_true, value_if_false)**) : When the logical expression is true, the value for "value_if_true" is displayed. When it is not true, the value for "value_if_false" is displayed.

<Example> Display "Pass" if the score is 70 or above, otherwise display "Fail"

Pass/Fail judgment
[Cell B2] =If(A2>=70, "Pass", "Fail")

	A	B
1	**Score**	**Pass/Fail judgement**
2	80	Pass
3		

[3] ([4]**=RAND()**) : Generates a random number between 0 and 1.

2 Simulation of Probabilistic Models

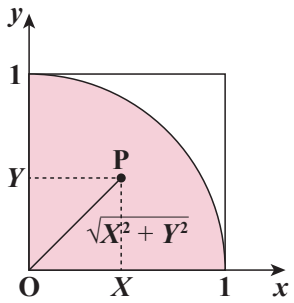
(1) ([5]**Monte Carlo method**) : A method for solving problems by using random numbers in probabilistic models.

(2) Simulation of the Approximate Value of Pi (π)

[1] Use the RAND function to randomly generate 100 points ($= N$) where both the X and Y coordinates are between 0 and 1.

[2] Count the number ($= n$) of ([6]**points that lie within a quadrant**).

[3] N: $n \approx 1$: $\frac{\pi}{4}$ Therefore, $\pi \approx \frac{4n}{N}$.



Points generated randomly using random numbers
[Cell D3] =B3^2+C3^2

Approximate value of pi
[Cell G3] =G3*4

	A	B	C	D	E	F	G	H
1	Simulation of approximate calculation of pi							
2	**Number of times**	**X-coordinate**	**Y-coordinate**	**The value of X^2+Y^2**	**Inside-outside determination**		**Probability of being inside a circle**	**Pi**
3	1	0.9066	0.9304	1.6876	0		0.800	3.200
4	2	0.2029	0.2804	0.3461	1			
5	3	0.2695	0.3481	0.4402	1			
6	4	0.7563	0.9880	1.2443	0			
7	5	0.5256	0.3030	0.6067	1			
8	6	0.1675	0.9127	0.9279	1			
9	7	0.4017	0.9968	1.0758	0			

Probability of a point being inside a circle
[Cell G3] =SUM(E3:E102)/100

Points generated randomly using random numbers
[Cell B3] =RAND()

Determines if a point is inside or outside a circle
[Cell E3] =IF(D3<=1,1,0)

*The accuracy of the approximation of the value of pi ([7]**increases**) as the number of points placed within the square increases.

(3) The Objective of the simulation:

[1] Simulate approximating π using random numbers, without measuring the circle's circumference and diameter.

[2] Draw random points inside a square, calculate the ratio of points inside a quarter circle

Warm Up

Using spreadsheet software, as shown in the diagram below, 100 points were plotted using random numbers between 0 and 1 for the X and Y coordinates. The number of points that fall within the area of a quadrant of a circle with a radius of 1 was examined, and from this, an approximate value of pi was calculated. Answer the following questions.

	A	B	C	D	E	F	G	H
1	**Simulation of approximate calculation of pi**							
2	**Number of times**	**X-coordinate**	**Y-coordinate**	**Distance from the origin**	**Inside-outside determination**		**Probability of being inside a circle**	**Pi**
3	1	0.9090	0.8882	1.6152	0		0.800	3.200
4	2	0.6769	0.1794	0.7003	1			
5	3	0.3127	0.9045	0.9570	1			
6	4	0.4324	0.0900	0.4417	1			
7	5	0.3476	0.8694	0.9363	1			
8	6	0.7042	0.9714	1.1998	0			
9	7	0.1034	0.0875	0.1354	1			

(1) Can this model be represented as a deterministic model or a probabilistic model?

(2) What is the term for the method that solves problems using random numbers in this way?

(3) The following formulas are entered in cells B3, E3, G3, and H3 in the diagram. Complete the following formulas by filling in the blanks **A** to **D** with the appropriate answers.

[Cell B3] = **A** ()

[Cell E3] = **B** (D3<=1, 1, 0)

[Cell G3] = SUM(E3:E102)/ **C**

[Cell H3] = **D** *4

(4) When 1,000 points were randomly plotted within a square, 750 of them fell inside a quadrant. Calculate the approximate value of pi in this situation.

Explanation

(1) Probabilistic model

(2) Monte Carlo method

(3) **A**: RAND **B**: IF **C**: 100 **D**: G3

(4) Since there are 1,000 points in the entire square and 750 of them are inside the quadrant, $\frac{750}{1{,}000} \times 4 = 3$
Therefore, the value of pi is 3

Try

1 A circle with a radius of 1 is drawn within a square with a side length of 2, with the circle centered at the origin. Random points were plotted within this square, and the number of points that fell inside the circle was counted. From this, the value of pi (π) was estimated. Answer the following questions.

(1) What is the term for the method that solves problems using random numbers?

(2) The procedure for calculating pi (π) based on (1) is as follows. Complete the following sentences by filling in the blanks **A** and **B** with the appropriate terms.

> [1] Place points randomly within a square.
> [2] Determine whether those points are (**A**) the circle.
> [3] Determine the (**B**) of the area of the circle to the area of the square using the number of points inside the circle and the total number of points within the square.
> [4] Use this (**B**) to calculate the value of pi (π).

(3) How does the accuracy of the estimated value of pi change as the number of points plotted within the square increases?

(4) When 10,000 points were randomly plotted inside a square, 8,000 of them were found to be inside a quadrant. Calculate the approximate value of pi in this situation.

2 Using spreadsheet software, as shown in the diagram below, 1,000 points were plotted using random numbers between 0 and 1 for the X and Y coordinates. The number of points that fall within the area of a quadrant of a circle with a radius of 1 was examined, and from this, an approximate value of pi was calculated. Answer the following questions.

	A	B	C	D	E	F	G	H
1	**Simulation of approximate calculation of pi**							
2	**Number of times**	**X-coordinate**	**Y-coordinate**	**Distance from the origin**	**Inside-outside determination**		**Probability of being inside a circle**	**Pi**
3	1	0.9090	0.8882	1.6152	0		0.800	3.200
4	2	0.6769	0.1794	0.7003	1			
5	3	0.3127	0.9045	0.9570	1			
6	4	0.4324	0.0900	0.4417	1			
7	5	0.3476	0.8694	0.9363	1			
8	6	0.7042	0.9714	1.1998	0			
9	7	0.1034	0.0875	0.1354	1			

(1) What is the term for the method that solves problems using random numbers in this way?

(2) The following formulas are entered in cells B3, E3, G3, and H3 in the diagram. Choose the formula that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

[Cell B3] = [1]

[Cell E3] = [2]

[Cell G3] = [3]

[Cell H3] = [4]

A G3*4 **B** IF(D3<=1, 1, 0)

C SUM(E3:E10002)/1000 **D** RAND()

Exercise

1 Answer the following questions.

(1) Regarding the IF function of spreadsheet software, choose the appropriate combination of terms to fill in blanks [1] to [3] from the options **A** to **D** below, and answer using the letter.

IF(([1]), ([2]), ([3]))

A [1] logical_expression [2] value_if_true [3] value_if_false

B [1] logical_expression [2] value_if_false [3] value_if_true

C [1] value_if_true [2] value_if_false [3] logical_expression

D [1] value_if_false [2] value_if_true [3] logical_expression

(2) Using spreadsheet software as shown in the figure on the right, you want to display a 1 in cell B3 if the score in cell A3 is 60 or above, and a 0 if it is not. Choose which function should be entered in cell B3 from the options **A** to **D** below, and answer using the letter.

	A	B
1	**Pass/Fail judgment**	
2	**Score**	**Pass/Fail judgement**
3	80	1

A [Cell B3] =IF(A3<=60, 1, 0)

B [Cell B3] =IF(A3<=60, 0, 1)

C [Cell B3] =IF(A3>=60, 1, 0)

D [Cell B3] =IF(A3>=60, 0, 1)